NANDITA KUMARI

Website |Google Scholar | GitHub | Contact

EDUCATION

- **PhD in Geosciences** , Stony Brook University GPA **3.97**/4.0 [2024]
- **B. E in Geoinformatics**, College of Engineering Guindy, Anna University, Chennai, GPA **8.93**/10 [2018]

AWARDS

- **NASA Exploration Science Forum** Travel Award, 2023
- **Zonta International Amelia Earhart Fellowship,** 2023
- **Featured Early Career Presentation**, Lunar Exploration Analysis Group (LEAG) Meeting 2022
- **LEAG Travel Award**, 2022,2023
- **Best Poster Award** in National Space Science Symposium 2019
- **Mike Barnsley Memorial Award**[1] by Alumni Association of CEG
- **Alumni Association Endowment Award 2007-2**[2] in Geoinformatics Engineering batch 2018
- **Merit Scholarship** from Army Welfare Association

INVITED TALK AND MEDIA INTERVIEWS

VELAY MEETING, STONY BROOK UNIVERSITY, 2023

CONTRIBUTIONS TO ARTEMIS III, SBU COVERAGE , LONG ISLAND MEDIA

AMELIA EARHART FELLOWSHIP, SBU COVERAGE

PROFESSIONAL EXPERIENCES & SERVICES

POSTDOCTORAL ASSOCIATE | *LUNASCOPE, BROWN UNIVERSITY 2024*

CHAIR | *NASA EXPLORATION SCIENCE FORUM 2020, 2023*

REVIEWER | *JGR:PLANETS, ICARUS, PSJ, GRL*

EXTERNAL REVIEWER | *NASA PANEL 2023*

EXECUTIVE SECRETARY | *NASA PANEL 2022*

MENTOR | *GWISE* [FALL'22-SPRING'23]

CHAIR | *LUNAR EXPLORATION ANALYSIS GROUP MEETING 2022*

SUMMER INTERN | *LUNAR AND PLANETARY INSTITUTE* [MAY'20-AUG'20]

VISITING STUDENT | *PHYSICAL RESEARCH LABORATORY* [MAY'18-JUL'19]

SUMMER INTERN | *PHYSICAL RESEARCH LABORATORY* [MAY'16-JUN'16]

PEER REVIEWED PUBLICATIONS

1. EXTENDED SILICIC VOLCANISM IN GRUITHUISEN REGION – REVISITING THE COMPOSITION AND THERMOPHYSICAL PROPERTIES OF GRUITHUISEN DOMES ON THE MOON(LINK)
2. EFFECTS OF SPACE WEATHERING ON THE CHRISTIANSEN FEATURE POSITION OF LUNAR SURFACE MATERIALS (LINK)
3. SURFACE CONDITIONS AND RESOURCE ACCESSIBILITY AT POTENTIAL ARTEMIS LANDING SITES 007 AND 011 (LINK)
4. GEOLOGICAL CHARACTERIZATION OF CHANDRAYAAN-2 LANDING SITE IN THE SOUTH POLE REGION OF THE MOON (LINK) [CO-AUTHORED]
5. THE SCIENTIFIC VALUE OF A SUSTAINED EXPLORATION PROGRAM AT THE ARISTARCHUS PLATEAU (LINK) [CO-AUTHORED]
6. TEMPERATURES OF THE LACUS MORTIS REGION OF THE MOON (LINK) [CO-AUTHORED]

[1] Annual award for highest marks in image processing in theory and lab

[2] Department rank 1 for the academic year 2016-2017

7. Dynamic Secondary Illumination in Permanent Shadows Within Artemis III Candidate Landing Regions (Link) [Co-Authored, PSJ]
8. Applications of Machine Learning to Spectral Data from Ongoing and Upcoming Lunar Missions [In Prep]

Non Peer Reviewed Publications

1. Space Weathering Across the Solar System: Lessons from the Moon and Outstanding Questions (Link) [Co-Authored]
2. Mission to Characterize Volatiles in Old, Cold, Permanently Shadowed Regions on the Moon (Link) [Co-Authored]

Conference Abstracts

1. Bulk Composition Characterization of The Lunar Surface Using Machine Learning: Applications to Ongoing and Upcoming Missions [NESF 2023]
2. Brewing Machine Learning Models For Accurate Analysis of Current and Future Datasets [LPSC 2023]
3. Procedure Standardization For Toolbox for Research and Exploration (TREX) Field Data Analysis [LPSC 2023]
4. Gruithuisen Domes: Silicic Volcanic Constructs on The Moon [LEAG 2022]
5. A Holistic View of Gruithuisen Domes – Silicic Constructs on the Moon [LPSC 2022]
6. Surface Illumination, Temperatures and Cold Traps at Two Potential Landing Sites Near the Lunar South Pole [NESF 2021]
7. Mid-Infrared Spectral Characteristics of Dust (Minerals and Meteorites) and Their Utility for Interpreting Remote-Sensing Data.[SOFIA Rock, dust and Ice Workshop 2021]
8. Episodic Volcanism and Fluvial Activities in a Floor-Fractured Crater at the Flanks of Arsia Mons [LPSC 2021]
9. Geological Diversity at Two Potential Landing Sites in the Lunar South Pole[LPSC 2021]
10. Artemis III EVA Opportunities on the Rim of de Gerlache Crater [NASA requested input for Artemis III Science Definition Team. Co-Authored]
11. Alternate Artemis III EVA Opportunities near de Gerlache Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
12. Artemis III EVA Opportunities on the Lunar Farside near Shackleton Crater [NASA requested input for Artemis III Science Definition Team, Co-Authored]
13. Different Trends of Variation in Christiansen Feature on The Lunar Surface: Effects of Iron Content and Particle Size [NESF 2020]
14. Investigation of Variation in Christiansen Feature With Respect To Albedo on the Moon Using the Diviner, M^3 and Kaguya Datasets [LPSC 2020]
15. Global Distribution and Characteristics of Domical Craters on the Moon [LPSC 2019]
16. Complex Impact Craters within Lunar Basins and Their Implication [NSSS 2019]

Field Work

Kilbourne and Hunts Hole, SSERVI (RISE4) 2022;2023

San Francisco Volcanic Field, SSERVI (CLSE) 2022

Utah Yellowcat Site, SSERVI(TREX),2022

Mount Ascutny, *New England NEIGC Field Trip 2019*

Technical Skills & Mission Experience

Programming	Python(Spatial and Non-spatial Data Analysis/Machine Learning, Proficient)
Software	ArcGIS, QGIS, ENVI, ERDAS Imagine, ISIS (Proficient) ; Davinci (Beginner)

DATA PROFICIENCY	Earth: LANDSAT[3] (VIS, THERMAL, PAN), ASTER Thermal InfraRed Data; Moon: LRO (WAC, NAC, WAC-DTM, Diviner, LOLA), Chandrayaan-1(M^3), Kaguya SELENE(MI, TC) ; Mars: MRO (CRISM, CTX, HiRISE), MGS(MOLA)
MISSION EXPERIENCE	LRO Diviner Science Team (2019-ongoing) ; Chandrayaan -2 landing site investigation science team (2018-2019)

LEADERSHIP SKILLS

SENATOR	| GRADUATE STUDENT ORGANIZATION OF SBU	[2021-2023]
STUDENT DIRECTOR	| *CEG TECH FORUM*	[2014-2018]
DIRECTOR	| *LITERARY CLUB OF ANNA UNIVERSITY*	[2018]
CORE MEMBER	| *SOCIETY FOR GEOINFORMATICS ENGINEERS*	[2014-2018]
EXPERIENCE MANAGER	| *AIESEC INTERNATIONAL, CHENNAI*	[JUL'15– JUL'16]

[3] LANDSAT Data from 5 to 8